METRIC

MIL-S-53085
30 August 1988

MILITARY SPECIFICATION

SURVEY INSTRUMENT AUTOMATED, INTEGRATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, medium-range, of 2km through 7km, electro-optical distance and angle measuring equipment. Distances are measured using diode-emitted infrared modulated light waves. Angles are determined by electronic measurement. AISI measuring unit will be solid-state construction and complete with accessories for unsupported operation.

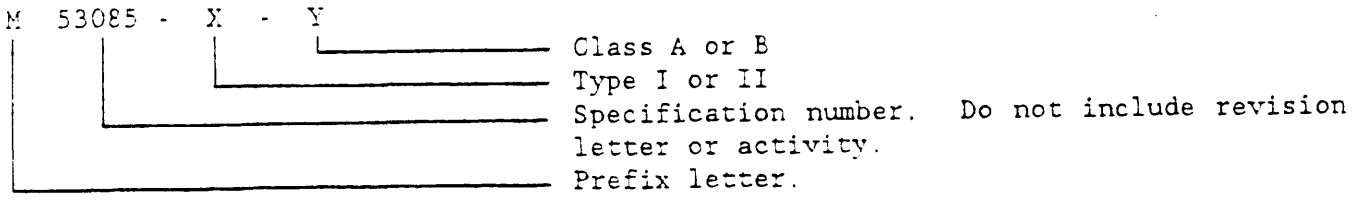
1.2 Classification. The AISI shall be of the following types, as specified:

Type I - One Tenth (00.1") second AISI with EDM modular unit of 7 kilometer capability.

Type II - One Tenth (00.1") second AISI with EDM modular unit of 2 kilometer capability.

Class A - With winterization kit.
Class B - Without winterization kit.

1.2.1 Part number. The military specification part number for the automated integrated survey instrument shall be as follows:



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, Attn: (STRBE-DS), Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-208	- Ink, Marking, Stencil, Opaque.
V-T-276	- Thread, Cotton.
W-C-440	- Clip, Electrical, General Specification for.
TT-F-325	- Filler, Engraving, Stamped Marking.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Process as for Cotton Cloth.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-795	- Cushioning Material (Flexible, Cellular, Plastic Film) For Packaging Applications.
PPP-C-1842	- Cushioning Material, Plastic, Open Cell.

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MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-H-639	- Hook (For Belts, Straps, and Equipage Items).
MIL-C-675	- Coating of Glass Optical Elements (Antireflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-T-3530	- Thread and Twine, Mildew Resistant or Water Repellent Treated.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.

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MIL-T-14132	- Theodolites, Surveying, Direction: Second Order.
MIL-R-14328	- Rubber Sheet, Synthetic, Medium Soft, General-Purpose Gasket Material (For Extreme Climatic Conditions).
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-B-23272	- Battery, Storage: Alkaline, Nickel-Cadmium.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-43566	- Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors.
MIL-T-52114	- Theodolite, Surveying, Direction: Third Order.
MIL-B-81705	- Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-785	- Reliability Program For Systems and Equipment Development and Production.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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PUBLICATIONS

TB MED-524 - Occupational Environmental Health Control of Hazard to Health from Laser Radiation.

(Application of copies should be addressed to the Commander, U.S. Army Publication Distribution Center, 1655 Woodson Road, St. Louis, MO 63114.)

NATIONAL BUREAU OF STANDARDS (NBS)

Special publication 374 - Method for determining the resolving power of photographic lenses.

(Application for copies should be addressed to the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.)

NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION (NOAA)

Standards and specifications for Geodetic Control networks, September 1984.

(Application for copies should be addressed to the National Geodetic Information Branch, N/CG174 Rockwall Bldg, Room 24, National Geodetic Survey (NOAA), Rockville, MD 20852.)

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. A surveying total station system, more specifically the automated integrated survey instrument, hereinafter referred to as AISI, shall combine three major modular subassemblies into a single, compact field survey instrument package: (1) the Electronic Theodolite (ELT), an auto-reading/recording digital theodolite with electronic display/control panel; (2) the Electronic Distance Meter (EDM), a low power, miniaturized distance measuring device with a range and accuracy compatible with the ELT; (3) the Data Recorder Unit (DRU), an automatic data recording/storage/processing terminal, ruggedized for field use with the ELT and EDM, and interoperable with any microprocessor through a RS232 interface. The AISI shall be mountable for operation on the standard military tripod specified in MIL-T-14091 using the universal tribrach specified in MIL-T-14132. The AISI shall also include: transport carry case or cases, retro-directive prisms, two 12 volt nickel cadmium (NiCad) battery packs, battery charger, auxiliary external power cable not less than 6 meters in length, tribrachs, accessories, tools, accessory cases and survey system software programs (see 6.2m) for computation, plotting or printing. When specified (see 6.2), a winterization kit, thermometer, altimeter-barometer, meteorological correction data, tripods, and backpack shall also be included. The AISI shall electronically measure the value of subtended horizontal and vertical angles, measure slope distances to a maximum range of 7km or 2km for a type I or type II instrument respectively and automatically record these data in a data storage/processing module for on-site reduction or transfer via computer interface to a microprocessor. Additionally, provided with appropriate input data, the AISI shall be able to cue the lay-out of horizontal and vertical earthmoving operations.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.7). Any changes or deviations of the AISI from the approved first article during production will be subject to approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish the AISI conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The AISI shall be fabricated from compatible materials inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the AISI may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste or reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the AISI may be newly fabricated from recovered materials to the maximum extent practicable, provided the AISI produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the AISI.

3.4 Environmental requirements.

3.4.1 Operational temperature. The AISI shall operate with no external assistance, without damage from an internal or external 12-volt power source, with the accuracy specified herein when in thermal equilibrium in air at any temperature from 50 to -20 °C. The AISI shall operate without damage and with the accuracy specified herein from -17 to -42 °C using an external winterization kit.

3.4.2 Storage temperature. The AISI shall not be damaged by storage under the following conditions:

High air temperature - 4 hours at +71 °C.
Low air temperature - 6 hours at -45 °C.

3.4.3 Solar radiation. The AISI shall operate without damage within the accuracy specified herein when subjected to solar radiation of 355 Btu/ft squared/hour, for not less than 4 hours in an ambient air temperature of 43 +0, -1.7 °C.

3.4.4 Humidity. The AISI shall not be damaged and shall operate within the accuracy specified herein when subjected to relative humidities of at least 95 percent at air temperature varying between -20 ±1, and 50 ±1 °C.

3.4.5 Rain. The AISI shall not be damaged nor show evidence of water penetration and shall operate within the accuracy specified herein before and after exposure to windblown rainfall of 7.62 ±2.54 cm/hr at wind speed not less than 48 km/hr for a period of not less than 60 minutes.

3.4.6 Vibration. The AISI in its transport carry case shall not be damaged by vibration when subjected to the vibration test specified in 4.5.2.7.5.1.

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3.4.7 Shock. The AISI in its transport carry case shall not be damaged when subjected to the shock test specified in 4.5.2.7.4.

3.4.8 Magnetic environment. AISI shall operate and measure distance within the accuracy specified in 3.15.2.1 and 3.15.3 and time specified in 3.15.2.2 while in the presence of a magnetic field of 3 gauss. The magnetic field orientation shall be east-west, north-south, and vertical.

3.4.9 Sand and dust. When the AISI and transport case are subjected to blowing sand and dust, the AISI shall show no evidence of sand and dust penetration into the optics of electronics package, or any lens abrasion. The transport case shall show no evidence of sand or dust penetration.

3.5 Reliability. The specified mean cycles between failure shall be 240 missions when tested as specified in 4.5.2.10.

3.6 Safety. AISI shall present no uncontrolled hazards to operators, bystanders, or maintenance personnel, and shall conform to all applicable requirements of TB MED 524 and requirement 1 of MIL-STD-454. AISI shall meet either the power and irradiance requirement of TB MED 524 for a class I device, or be less than the irradiance criteria for a class I device and be less than five times the power limitations for a class I device (see 6.6 for safety ordering data).

3.7 Human factors engineering. The AISI shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), design for maintainability (5.9), small systems and equipment (5.11), hazards and safety (5.13), and user-computer interface (5.15) of MIL-STD-1472, as applicable.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, and test equipment (see 6.3 and 6.4) shall be adequate to perform the intended function of assisting or conducting maintenance operations during testing as specified in 4.5.2.11.

3.8.2 Maintenance ratio. AISI shall have a maintenance ratio of not more than 0.014 when tested as specified in 4.5.2.11.2. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total missions performed. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8.3 Ease of maintenance. AISI construction shall provide ease of access to batteries and fuses (or reset buttons) such that tools are not required to connect a battery or external power nor to change a fuse. AISI construction

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shall provide a self-test display to verify satisfactory operating capability prior to initial measurement. Design for maintainability shall comply with the provisions of 3.7.

3.9 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the AISI shall conform to MIL-STD-461, methods RE02 for radiated and CE03 for conducted tests, for class B equipment. The frequency spectrum scan shall be from 0.014 through 1 GHz during the radiated tests and from 0.020 through 50 MHz during the conducted tests with the following exceptions: (a) Broadband radiated emissions (RE02) at 0.3 MHz and 0.7 MHz shall be increased 5 dB and, (b) Broadband radiated emissions (RE02) at 150 MHz shall be increased 6 dB above the requirements of MIL-STD-441.

3.10 Physical characteristics.

3.10.1 Portability. AISI shall be furnished in soldier portable packages.

3.10.2 Weight. The basic equipment weight (ELT, EDM, DRU, tribrach and NiCad battery exclusive of tripods but including primary power source) shall be not more than 25 pounds. The transport weight (basic equipment and transport carry case) shall be not more than 40 pounds.

3.10.3 Electronic theodolite (ELT). The ELT is the electronic horizontal and vertical angle measuring component of the AISI and will automatically display the observed angle at a precision and repeatability required to satisfy 2d order class II accuracy conforming to, NOAA Publication (Standards and Specifications for Geodetic Control Networks, September 1984).

3.10.3.1 Alidade assembly. The alidade assembly shall include the alidade proper and the lower motion. The alidade proper shall include altitude levels on a self-indexing device, the telescope producing an erect image when sighted, the horizontal axis, and wiring for electric illumination. The lower motion shall include the vertical axis, an exterior upper and lower circle graduated in degrees, provision for electric illumination, and a means for attaching to the tribrach.

3.10.3.1.1 Telescope. The telescope shall consist of the optical system, sights or a reticle for aiming, and a retractable or removable sunshade. The overall length of the telescope shall be not greater than 180 millimeters (mm) when the diopter scale is set to zero. A means for illuminating the reticle shall be internally mounted in the telescope, and a means for controlling the illumination shall be provided. The centering of the telescope over the vertical axis shall be such that the combined error and the change in the line of collimation resulting from the travel of the negative lens over the focusing range shall not change the line of collimation by more than 10 seconds of arc for type I and II AISI instruments over a focusing range from 4.6 meters to infinity. The telescope shall balance when focused at infinity with the sunshade in the extended position.

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3.10.3.1.1.1 Optical properties. The optical properties shall conform to the following requirement:

Magnification, telescope	- 28 diameters, plus or minus 1 diameter
Effective aperture of objective, (min)	- 40 mm
Field of view, (min)	- 1 degree 20 minutes
Focus, minimum, not greater than	- 2.2 meters
Resolution	- 4 seconds or better of arc

The principal axis of the lens system shall be in alignment and coincident with the collimation of the telescope in adjustment. A means of collimating the telescope line of sight with the electronic line of sight shall be provided. The field of view shall be substantially flat from the center to a zone outside the stadia lines equal to 1/4 of the stadia interval, and shall show no marked curvature from there to the edge of the field. Except for the registered face of the reticle, the internal lens-air surfaces of the telescope shall be coated with a reflection-reducing film in accordance with MIL-C-675.

3.10.3.1.1.2 Reticle. Unless otherwise specified, the reticle shall consist of a pattern of opaque lines similar to that shown in figure 1, registered on optical glass. The single vertical crossline shall subtend equal angles with the adjacent double lines within plus or minus 0.6 seconds. The width of the lines shall be 2.5 to 4.0 microns. The double vertical lines shall subtend an apparent angle of 40 seconds, plus 5 seconds or minus 10 seconds. The horizontal and vertical stadia lines shall be so spaced that 1/100 of the distance from a stadia rod at a known distance greater than 30 meters from the instrument shall be subtended with an error in reading not to exceed 0.3 percent. The lines shall subtend 5 to 7 minutes of arc in length and shall be parallel to the corresponding crossline. The reticle shall be mounted in a metal cell which shall not become distorted during adjustment for collimation. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors of the lines of sight shall be not more than 6 seconds.

3.10.3.1.1.3 Telescope eyepiece. The eyepiece shall be so designed that the operator can place his eye in a position coincident with the exit pupil. The eyepiece shall be equipped with a diopter scale marked from zero in plus and minus units, and shall have an index mark for referencing the setting.

3.10.3.1.2 Standards. The standards may be adjustable for height or may be factory adjusted and set permanently. The horizontal axis of the telescope shall be so nearly perpendicular to the vertical axis of the AISI that the error due to the difference in the height of the standards shall be not more than 12 seconds.

3.10.3.1.3 Horizontal and vertical axes. Bearings shall be of the free-movement type with a motion clamp and a tangent screw for slow motion control and precise adjustment. The horizontal plane shall not show signs of backlash, binding, and endplay. Tangent screws shall operate without binding, backlash,

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and endplay. All bearings, motion clamps and screws shall be protected against moisture, dust, and deleterious foreign matter; the bearings, motion clamps and screws need not be sealed.

3.10.3.1.4 Circles.

3.10.3.1.4.1 Horizontal and vertical circle. The horizontal and vertical circles shall be graduated with encoded tracks electronically readable by the integral microprocessor. Sensing of diametrically opposite projections shall provide for a means of automated correction for circle eccentricity and variations in circle graduations. In conjunction with an electronic micrometer, the resolution of the circle readings shall be 0.5". The initial display time shall not exceed 1.0 second. Inclination of the vertical axis shall be detected, corrected and automatically applied to readings by a compensator system. Vertical angle measurements shall be automatically indexed within a sensing range of +/- 10' to an accuracy of +/- 0.5". The units of angular measurement shall be operator selectable on the keyboard in either sexagesimal or decimal.

3.10.3.1.4.2 Setting circles. A horizontal and vertical circle setting device shall be furnished for setting the circles in various positions of an observation program. The circles will be provided with a means of selecting the units of angular measure so they will correspond to those selected on the microprocessor. The circle setting device shall have free movement without binding throughout full revolution. Positive means shall be provided to prevent inadvertent movement by the observer.

3.10.3.1.5 Microprocessor display/control panel.

3.10.3.1.5.1 Keyboard. Panel dials, displays, knobs and switches shall be functionally grouped, color coded and constructed for simple operation located on both sides of the theodolite. Each shall be marked to identify its functional name, symbol, operating instruction reference. To assure reliable operation in adverse field conditions, the keyboard will be weatherproof through the use of gasketed assembly, silicon-sealed screw heads, molded in display windows and waterproof keyboard buttons. Keyboard controls shall be able but not limited to initiating the microprocessor functions listed in 3.10.3.1.5.4. Control and display shall comply with the requirements of 3.7.

3.10.3.1.5.2 Displays. Light emitting diodes or illuminated liquid crystal technology shall be used for data displays. Displays shall be visible in bright sunlight.

3.10.3.1.5.3 Illumination. The electronic control panel shall have internally mounted illumination for night operation. Panel dials, knobs and switches shall be functionally grouped and constructed for simple operation. All external lighting shall be a blue-green color between 400-700 nanometers in wave length with no light visibility beyond a 50 meter circle with the instrument as center point under moonless and starless night conditions. All external visible light produced shall be intensity adjustable.

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3.10.3.1.5.4 Microprocessor. The AISI microprocessor shall have a non-volatile memory of adequate capacity to store all instrument constants and operating parameters and random access program memory with enough capacity to perform simple data reductions. The microprocessor shall provide initiation and control of a system self-test. The test shall include but is not limited to; power source check, EDM operational check, and display test. The microprocessor shall be capable also of executing the following listed functions by means of single or multiple keystroke entries from the control panel keyboard:

OPERATIONAL COMMANDS
Power on-off
Illumination on-off
System Self-test
Clear entries/commands
Select/display units of angular/distance measure
Select precision of angular display readings
Set horizontal circle orientation
Measure/display/record/store horizontal angle or direction
Measure/display/record/store vertical or zenith angle
Measure/display/record/store horizontal collimation error
Measure/display/record/store inclination of standing axis
Level standing axis
Measure/display/record/store slope or horizontal distance
Measure/display/record/store height differences or elevations
Display coordinates
Display stake-out direction/distance
Display error/system messages

DATA ENTRY COMMANDS
Enter vertical circle index error
Enter horizontal collimation error
Enter stake-out data
Enter initializing horizontal/vertical datum
Enter height of instrument/target
Enter instrument/target constants
Enter scale corrections
Enter atmospheric data/corrections
Enter system codes (stationing, record)
Enter system format parameters (baud rate, parity) for data transmissions
Enter instrument/target eccentricity

3.10.3.1.5.5 Internal electronics design. The AISI's internal electronics package shall be constructed in modular units and shall have a positive guide arrangement to facilitate removal and replacement of the module. Test points for use in internal electronic component/circuitry diagnostics shall be easily identifiable and accessible. The electrical circuitry, including all components and connections, except as specified below, shall be protected with a varnish

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conforming to MIL-V-173, composition I or II, as applicable, with 1 percent copper 8-quinolinolate (by weight) base on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management Control District.

3.10.3.1.6 Axes motion controls. The horizontal and vertical axes shall be suspended or supported on bushings or bearings. Bearings shall be of the free-movement type with a motion clamp and a tangent screw for slow motion control and precise adjustment. Movement through the vertical tilt limits, minimum ±45 degrees, and the horizontal plane shall not show signs of backlash, binding, o endplay. Tangent screws shall operate without binding, backlash, or endplay. All bearings, motion clamps and screws shall be protected against moisture, dust, and deleterious foreign matter; the bearings, motion clamps and screws need not be sealed. The horizontal-motion tangent screw and the vertical slow-motion screw shall rotate the telescope not less than 6'45" nor more than 20'15" per revolution. The horizontal and vertical motion tangent screws shall be long enough to produce at least 3 degrees of travel. Control knobs and leveling screws shall have plastic or plastic coated heads or when specified in 6.2 a winterization kit shall be provided. Horizontal and vertical motion controls shall be two-speed and coaxial with motion locks to enhance rapid and precise sighting. Tangent screws shall have a takeup for wear adjustment and shall be removable for cleaning.

3.10.3.1.7 Optical plummet. The optical plummet shall be integral with the alidade or shall be attached to the tribrach and shall be adjustable. The reticle pattern shall divide the field of vision into four 90-degree quadrants. The optical system shall produce a clear image throughout the operating range from tripod height to not less than 110 feet (33.5 meters). The line of sight of the optical plummet shall be coincident with the vertical axis of the AISI. A means of illuminating the reticle shall be internally mounted in the plummet and a means of controlling the light intensity shall be provided.

3.10.3.1.8 Levels. The instrument shall be equipped with a plate level, circular level, altitude level or a self-indexing device, where applicable.

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3.10.3.1.8.1 Plate level. The plate level vial shall be graduated in 2 millimeter intervals and shall have a sensitivity of 20 seconds per 2 millimeter graduations, plus or minus 4 seconds. Graduations shall be filled with a paste conforming to TT-F-325, type II. Graduations shall be symmetrical about the exposed portion and shall extend far enough to include ends of the bubble at the operating temperature specified herein. The plate level shall be mounted as close as practicable to the vertical axis and shall be protected. A means of adjusting the level shall be provided.

3.10.3.1.8.2 Circular level. The instrument shall be equipped with a circular level. The circular level shall have a sensitivity of 10 minutes, plus or minus 3 minutes, for each 2 millimeters of radial distance. It shall be mounted on the alidade or on the tribrach.

3.10.3.1.9 Tribrach assembly. There shall be three tribrachs each consisting of the leveling screws, base plate, and a device for locking the alidade assembly. The base plate shall have at its center a 5/8-inch nut of the unified coarse thread series (5/8-11 UNC-2B), class 2B, 11 threads per inch, as defined in FED-STD-H28. The leveling screws shall permit leveling of the theodolite when the base plate is 6 degrees out of level. The design of the tribrach shall be such that removing and returning the alidade shall cause a change in the centering of the alidade not to exceed 0.05 mm. Tribrachs shall be interchangeable with all alidades procured under this specification and from the same manufacturer. The leveling screws shall have a takeup for wear adjustment and shall be removable for cleaning. If optical plummet is integral to tribrach (see 3.10.3.1.7).

3.10.3.2 Power. The ELT/EDM shall operate from an internal NiCad, 12 volt DC battery conforming to MIL-B-23272 or from an external lead-acid 12 volt DC battery conforming to MIL-B-11188. AISI shall contain built-in protection against reverse polarity from the power source, including battery compartment design to preclude improper battery insertion. AISI winterization kit heating elements shall operate from lead-acid 12 volts DC battery.

3.10.3.2.1 System power. The ELT/EDM shall be provided with two NiCad battery packs with a combination of cells, yielding 12 volts DC operation, supporting a minimum of 500 angle measurements. The battery shall be rechargeable (see 3.10.3.2.3), supplied with the system and shall provide at least one hour of measuring time at 20 ^{o}C. The battery or batteries shall be spillproof and shall have no detectable liquid or gas regardless of operation or storage orientation and shall not be damaged, when subjected to shock testing as specified in 4.5.2.7.4.

3.10.3.2.2 Auxiliary power. The ELT/EDM shall also operate using lead-acid storage batteries conforming to MIL-B-11188, 12-volt dc, and providing the same operating capability as the internal power source. The external power cable provided with the system shall not be less than 6 meters in length and shall terminate in battery clips conforming to W-C-440, type PC, with insulators.

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3.10.3.2.3 Battery charger. System batteries shall be rechargeable using 110/220 volt, 50/60 Hz charger supplied with the system. The battery charger shall be capable of concurrently charging at least two 12-volt/2 ampere hour batteries or a built-in battery charger with each NiCad battery pack shall be furnished. The battery shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

3.10.4 Electronic distance meter (EDM). The EDM shall be modular in design and shall be mountable on the telescope of the ELT with minimal distance between the optical centerlines of the telescope and EDM optics. The design of the telescope mount shall allow adjusting for parallelism between the ELT and EDM optical line-of-sight. The mounted EDM shall be completely stable in all telescope positions, without assistance of counterbalance springs or external weights when the ELT's vertical motion lock is fully released. Distance measurements shall be acquired in either a normal mode (individual measurements manually cued and displayed) or tracking mode, (continuously measured and displayed). The EDM shall be capable of operating from the power supplied through the ELT mount, or using power cable directly from a separate external power source (see 3.10.3.2). Mounting on a conventional theodolite shall be possible using an optional mounting bracket and EDM integral keyboard for data entry and operational commands. The keyboard control on the EDM shall allow for the entry of the following commands: Power/illumination on-off; Measure and display distance; Activate tracking mode; Stop tracking mode; Self-test (signal strength, power level), Data entry (prism constant, scale correction). The units of distance measurement and data entry shall be keyboard selectable in either th Metric or English system. The ELT keyboard shall provide the above functions an. in addition, extend the capabilities of the ELT/EDM combination by providing for remote EDM control, horizontal distance display, height difference, differential ground point stake-out, coordinate differences, and atmospheric corrections. Signal return strength shall be automatically attenuated to insure unambiguous readings throughout the specified ranges of the EDM.

3.10.4.1 Power. Power shall conform to 3.10.3.2.

3.10.4.2 System power. System power shall conform to 3.10.3.2.1.

3.10.4.3 Auxiliary power. Auxiliary power shall conform to 3.10.3.2.2.

3.10.4.4 Battery charger. Battery charger shall conform to 3.10.3.2.3.

3.10.5 Data recording unit (DRU). This modular component of the AISI shall automatically acquire data from the ELT and down-load through an appropriate interface to an office data processing unit. Additionally, it shall be operator programmable to perform data storage, transfer and customized computing tasks. Program language shall be user friendly such as BASIC. Prepared programs shall be supplied to accomplish field routines such as coordinate geometry calculations, stakeout, precise leveling, construction alignment. The DRU shall be mountable to standard military tripods. A means of reference to frequently entered field data codes, DRU commands, or program instructions shall be

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attachable to the DRU for operator convenience. DRU memory shall be divided into data memory and program memory thus providing a means of applying, storing and accessing field data independent of program memory. Non-volatile memory capacity shall be adequate to store data transfer parameters (baud rate, parity code) and field station data (horizontal and vertical angle, horizontal distance, and station designation/information code) for 500 occupied survey stations. Power source necessary to maintain memory non-volatility shall be internal, independent of operation power and have a minimum supporting life of 12 months.

3.10.5.1 Keyboard. The basic criteria on features shall conform to 3.10.3.1.5.1.

3.10.5.2 Data structure. The DRU shall be capable of recording, accessing, editing and executing program files. The data organization shall allow separate storage and retrieval of data files containing station identification information, horizontal angle, horizontal distance and vertical angle parameters for 500 ground control points. The design, size and number of both data and program files shall be sufficient to support field survey data collection, encoding and program commands. The DRU shall be capable of data transfer using interfaces such as RS232. Baud rate shall be a selectable transmission range of 110 to 9600 Baud. Additional data transfer information such as parity, character format, protocol and terminal operation selectables shall be available at the DRU.

3.10.5.3 Display. Light emitting diodes or illuminated liquid crystal technology shall be used for data displays. Multiple or single displays shall be capable of comprehensive presentation of operator prompts, program instructions and data. Display shall be visible in bright sunlight.

3.10.5.4 Power. The DRU shall contain an internal rechargeable power supply. Operating power loss shall not affect any data stored in the non-volatile memory and shall be preceded by a display warning.

3.10.6 Reflector set. Two retro-reflector prism sets shall be provided as a component of the AISI. Each set shall consist of:

a. Sufficient number of prisms to satisfy maximum range capabilities for type I and type II AISI's (see 3.15.3.2).
b. Single and multi-prism holders to accommodate all prisms provided.
c. Prism adapter to allow mounting of a combination of prism holders.
d. A mounting adapter for mounting both the prism holders and adapter in the universal tribrach specified in MIL-T-14132. The mounting adapter shall permit the prisms to be rotated through 360 degrees of arc about the vertical axis without disturbing either the tribrach or plumb of the assembly over the survey mark. A clamping device of friction fit shall be provided for locking the prisms at any point after rotation.
e. A transport carry case or cases conforming to 3.10.7.

3.10.7 Transport carry case. The transport carry case (housing the electro-optical package, accessories and tools) shall be metal or reinforced plastic

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doors. Grab handles shall be provided

laminate, or molded plastic designed with reinforced ribs, and shall have removable inner liner which is molded to accept the various AISI accessories. The case shall withstand operating, transport, and storage usage and shall be adequately blocked and padded to prevent damage that might otherwise result if the transport case is dropped from a distance of 91 cm, and to protect the instrument against damage, during field transport, shipment, and storage, from rainfall, shock, and vibration as specified herein. Each case shall contain a desiccant. The case shall open in such a way that hardware will not fall out when the case is opened. The case shall be provided with heavy-duty locking catches and, where possible, shall provide means for padlocking. All items listed in 3.10.9 (except the backpack) shall be stored in the transport carry case. The cases shall be the same color as specified for the instrument (see 3.13) and shall be no larger than 63.5 cm x 30.5 cm x 25.4 cm. Each transport case shall be labelled with an identification plate and loading plan (see 3.11) which indicates the contents and shall weigh no more than 15 pounds.

3.10.8 Accessories and tools. The following shall be the minimum furnished with each AISI as accessories and tools:

Item	Quantity
Spare Fuses	4 each
Brush, lens, camel hair	1 each
Tissue, Lens, pkg.	1 each
Operator Manual	1 each
Covers, lens, ELT, (both ends)	1 set
Cover, lens, EDM	1 set
Cover, Instrument, waterproof	1 each
Pole, plumbing; 2.15 meters (adjustable), with circular bubble	2 each

The contractor may add items to this list as required to perform operator maintenance and field adjustments. Accessories and tools shall be provided with a separate case similar in construction to type indicated in 3.10.7, or packed with major components in their respective transport cases.

3.10.9 Optional equipment. The following equipment is optional and is available with the AISI when specified by ordering data (see 6.2):

	Item	Quantity
a.	Backpack (see 3.10.9.1)	2
b.	Winterization kit (see 3.16.2 - 3.16.5)	1
c.	Tripod (see 3.10.9.6)	3
d.	Thermometer, centigrade (commercial std) (see 6.2)	1
e.	Altimeter/barometer, (mm Hg graduation) commercial std with meteorological correction charts (see 6.2)	1 2
f.	Adapter 110/220v, 50/60 Hz. (use with battery charger)	1

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3.10.9.1 Backpack. When specified (see 6.2), a means for transporting AISI by backpacking shall be provided. The means for backpacking shall comply with the requirements of 3.7. Canvas, if used, shall conform to CCC-C-419, type and weight to suit the purpose, and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f, subdued or camouflaged color. The backpack shall be easy to use and shall be comfortable to wear for male and female personnel ranging anthropometically from the 5th to 95th percentile.

3.10.9.2 Winterization kit. When specified (see 6.2) a winterization kit shall be furnished for use with AISI at temperatures below -20 °C. The winterization kit shall include covers for all control knobs and leveling screws, shields for the telescope, the microscope, and the optical plummet eyepieces. The kit shall be contained in a cotton-duck carrying case. The kit shall be capable of heating AISI to operating temperature from -32 °C within 25 minutes and from -43 °C within 45 minutes. An auxiliary power source (see 3.10.3.2.2) can be used.

3.10.9.2.1 Covers. Covers for the AISI, control knobs, and leveling screws shall be made of rubber conforming to MIL-R-14328. Covers need not be furnished for control knobs and leveling screws with plastic or plastic coated heads.

3.10.9.2.2 Shields. Shields for eyepieces shall be laminated fabric and shall not crack or split at the low operating temperatures specified herein when being attached or removed. The shields shall fit snugly over the eyepiece without disturbing the focus or decreasing the field of view.

3.10.9.2.3 Case. The case shall be of cotton duck and shall contain all the components of the winterization kit. The case shall have a slide fastener, a belt loop, and a double belt hook conforming to MIL-H-639, type I, class 1. The cotton duck used shall conform to CCC-C-419, type I, No. 12, hard-texture and rendered fungus-resistant and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f, and dyed olive drab 7. Webbing used shall conform to MIL-W-530, type IIa, class 4, dyed olive drab 7. Tape shall conform to MIL-T-43566, type I, class 4. Thread used in fabricating the case shall conform to V-T-276, type III B, ticket No. 10/4, dyed olive drab 7, and rendered fungus-resistant in accordance with MIL-T-3530, type I, class 1.

3.10.9.3 Tripod. When specified (see 6.2), three tripods conforming to MIL-T-14091 shall be furnished.

3.11 Identification marking. The AISI shall be identified in accordance with MIL-STD-130. The marking shall be applied on plots conforming to MIL-P-514, type I, style 1, composition c, of type I grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible. In addition to the information required, the plate shall be marked with the following:

US
NSN

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A loading plan shall be ink marked using stenciling on the top inside cover of transport carry case conforming to A-A-208. When specified (see 3.10.9.1) canvas case and the backpack shall be identified by stenciling using ink conforming to A-A-208.

3.12 Interchangeability. All parts of the same manufacturer having the same system part number or function shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.13 Treatment and painting. The portions of the AISI, cases and accessories normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G. Unless otherwise specified (see 6.2), the color shall be Green 383.

3.14 Lubrication. The AISI shall be lubricated prior to assembly with grease conforming to MIL-G-23827. The grease shall permit the theodolite to operate throughout the specified temperature ranges.

3.15 Performance.

3.15.1 Calibration. Each AISI shall be calibrated by the contractor so that there shall be no instrument constant or retrodirective prism offsets that cannot be corrected in the instrument prior to distance and angle measurement. If such constants or effects do exist, they shall be stamped on the applicable hardware in a conspicuous place to facilitate presetting AISI. A means shall be provided for adjusting AISI to the proper constant without use of special tools.

3.15.2 Electronic theodolite.

3.15.2.1 Measurement accuracy. The measurement of horizontal and vertical angles shall yield a Standard Deviation of mean pointings between direct and reverse sightings of 0.8" or less. Maximum resolution of displayed angular values shall be 0.1".

3.15.2.2 ELT and EDM measurement time. After setup and actuation of the AISI's off-on power switch to the "ON" position, the time required for an initial distance and angular reading in ambient temperatures from 49 to -20 °C shall be not more than 4 minutes. Subsequent distance and angular readings shall be accomplished in not more than 6 seconds. The same time limitation shall apply to AISI after having been warmed from -43 to -20 °C by the winterization kit, when applicable, as specified in 3.10.9.2.

3.15.2.3 Data processing, storage, transfer. Field measurement data shall be automatically and manually manipulated in accordance with 3.10.3.1.5.4 without loss of information or identification.

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3.15.3 Electronic distance meter.

3.15.3.1 Measurement accuracy, type I and type II. The maximum allowable mean-square error for a mission of 10 observations at an ambient temperature range of -5 to +50 °C shall not exceed ±5mm+5ppm. (Note: Temperatures shall be accurate to ±1 °C and pressures to ±5mm of mercury.)

3.15.3.2 Range. Atmospheric conditions and the number of prisms used as targets determine the range capabilities of the EDM. Minimum range capability under all atmospheric conditions shall not exceed 2 meters. The following maximum ranges shall be achieved at the accuracy specified in 3.15.3.1 under the conditions stated for type I and type II AISIs.

Table I. Atmospheric Conditions*.

	Poor	Average	Excellent
	Type I.		
1 Prism	1.0Km	2.0Km	2.5Km
3 Prism	1.2Km	3.5Km	5.0Km
7 Prism	1.7Km	4.5Km	6.0Km
11 Prism	1.8Km	5.0Km	7.0Km
	Type II.		
1-3 Prism	0.5Km	1.5Km	2.5Km

*Atmospheric Conditions are defined as:

Poor - Visibility 3Km with strong haze, bright sun, and heat radiation.
Average - Visibility 15-20Km, light haze, moderate sun, light heat radiation.
Excellent - Visibility 40Km, clear, overcast to partly cloudy, no heat radiation.

3.15.3.3 Measurement time. Measurement time shall conform to 3.15.2.2.

3.16 AISI system software. The system software shall be a fully integrated, three dimensional, ground modeling, design, and drafting system. The system should include such applications as field note reduction, automated data collection, traverse balancing, surface modeling, contour mapping, coordinate geometry, roadway design, drainage design, earthwork volumes by average-end-area or true prismoidal calculations, and three dimensional projections. The software shall be executable with MS-DOS type system and shall run on IBM and IBM compatible hardware. The software may contain separate packages to perform functions such as drafting and design calculations. However, maximum integration into one package is highly desirable. Software application will be in the two

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areas of topographic (control) survey and construction survey. In both areas the software will be required to transfer data between recorder and processor and to the peripheral equipment such as printer and plotter. Representative tasks to be performed in the topographic area include: storage of raw field data; compute, adjust, plot and annotate three dimensional coordinates using coordinate geometry (includes traverse, intersection, and resection); compute a level line; compute astronomic azimuth (stellar); coordinate conversion universal transverse mercator (UTM) to geographic, world geodetic system (WGS), state plane coordinate conversion; zone to zone conversion (UTM); and compute convergence of coordinate lines. In the construction survey task area the following representative tasks are critical: storage of raw field data; compute, adjust, plot, and annotate three dimensional coordinates using coordinate geometry (includes traverse, intersection and resection); compute and adjust level data; compute and format for field location, road, airfield, buildings, and utilities (centerline, grade, earthwork); compute and plot profiles, cross sections, vertical curves, and horizontal curves; compute and plot preliminary/final road, airfield and architectural construction plans. Software must be interactive, menu driven, and user friendly to the maximum extent possible, and shall comply with the user-computer interface requirements of 3.7.

3.17 Workmanship. All parts, components, and assemblies of the AISI instrument, tripod, and cases including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful extraneous material. All parts shall be finished. There shall be no cracks, sharp bends, dents, or chips in finished parts. There shall be no loose solder or insecure electrical connections. The fit of assemblies and subassemblies shall be made without imposing strain or forcing parts. All threads requiring movement and all sliding and rotatable parts shall move without binding, backlash, or endplay. Clamping actions, when required, shall be positive and, when unclamped, shall not cause drag. No lateral motion shall exist between the electro-optics housing and support. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between glass and coated film, or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Bearing surfaces shall show no scratches, or any irregular surface blemishes. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required. Focusing components or assemblies shall retain their focus when finger pressure is released and the telescope shall not change focus when transited. Metal surfacings on the optical components for all prisms and lenses shall not shift. There shall be no flickering of the illumination system when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall show no scratches or surface blemishes. Mirrors shall not be tarnished, and backing shall be uniform and unspotted. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required.

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4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection/test. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. The AISI and associated (optional) equipment shall be subjected to the tests marked "X" in column 2 of table III. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The AISI component, parts, and associated (optional) equipment shall be examined for the following defects.

Table II. Examination schedule.

Number	Defects	Requirement paragraph
Major		
101.	Any AISI component or part or associated (optional) equipment missing.	3.1
102.	Material not as specified.	3.3
103.	Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the AISI.	3.3.2
107.	Ease of maintenance not as specified.	3.8.3
108.	Weights not as specified.	3.10.2
109.	Alidade assembly not as specified.	3.10.3.1
110.	Telescope eyepiece not as specified.	3.10.3.1.1.3
111.	Horizontal and vertical axes not as specified.	3.10.3.1.3
112.	Keyboard not as specified.	3.10.3.1.5.1
113.	Data display illumination not as specified.	3.10.3.1.5.2
114.	Control panel illumination not as specified.	3.10.3.1.5.3
115.	Microprocessor not as specified.	3.10.3.1.5.4
116.	Internal electronics design not as specified.	3.10.3.1.5.5

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Table II. Examination schedule. (continued)

Number	Defects	Requirement paragraph
117.	Horizontal and vertical axes motion control not as specified.	3.10.3.1.6
118.	Levels not as specified.	3.10.3.1.8
119.	Battery charger not as specified.	3.10.3.2.3
120.	Electronic Distance Meter (EDM) not as specified.	3.10.4
121.	Data Recording Unit (DRU) not as specified.	3.10.5
122.	Reflector set not as specified.	3.10.6
123.	Transport carry case not as specified.	3.10.7
124.	Winterization kit performance not as specified.	3.10.9.2
125.	Kit control knobs and leveling screw covers not as specified.	3.10.9.2.1
126.	Kit eyepiece shields not as specified.	3.10.9.2.2
127.	Case for winterization kit not as specified.	3.10.9.2.3
128.	Prism not as specified.	3.15.3.2
129.	Workmanship not as specified.	3.17
Minor		
201.	AISI packages portability not as specified.	3.10.1
202.	Tools and accessories not as specified.	3.10.8
203.	Thermometer not as specified.	3.10.9.d
204.	Altimeter/barometer or meteorological correction charts not as specified.	3.10.9.e
205.	Adapter not as specified.	3.10.9.f
206.	Tribrach not as specified.	3.10.3.1.9
207.	NiCad battery pack not as specified.	3.10.3.2
208.	Backpack portability not as specified.	3.10.9.1
209.	Tripod not as specified.	3.10.9.3
210.	Identification or loading plan marking missing, incomplete or illegible.	3.11
211.	Parts interchangeability not as specified.	3.12
212.	Treatment and painting not as specified.	3.13
213.	Lubrication not as specified.	3.14

4.5.2 Tests.

4.5.2.1 Conditions of test. Prior to beginning the tests, the contractor shall prepare AISI to operate as specified herein.

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4.5.2.2 Test schedule.

Table III. Test.

First article	Quality conformance	Test	Test para.	Requirement para.
X		High-temperature operation	4.5.2.7.1.2	3.4.1
X		Low-temperature operation	4.5.2.7.1.5	3.4.1
X		High-temperature storage	4.5.2.7.1.3	3.4.2
X		Low-temperature storage	4.5.2.7.1.6	3.4.2
X		Solar radiation	4.5.2.7.1.4	3.4.3
X		Humidity	4.5.2.7.2	3.4.4
X		Rain	4.5.2.7.3	3.4.5
X		Vibration	4.5.2.7.5	3.4.6
X		Shock	4.5.2.7.4	3.4.7
X		Magnetic environment	4.5.2.7.6	3.4.8
X		Sand and dust	4.5.2.7.8	3.4.9
X		Reliability	4.5.2.10	3.5
X		Safety	4.5.2.9	3.6
X		Human engineering	4.5.2.8	3.7
X		Maintainability	4.5.2.11	3.8
X		Electromagnetic Interference	4.5.2.7.7	3.9
X		ELT subassembly system	4.5.2.3.2	3.10.3
X		Collimation	4.5.2.3.3	3.10.3.1.1
X		Telescope	4.5.2.3.4	3.10.3.1.1 3.10.3.1.1.1
X		Reticle lines	4.5.2.3.5	3.10.3.1.1.2
X		Height of standards	4.5.2.3.7	3.10.3.1.2
X		Vertical circle indexing and vertical collimation	4.5.2.3.6	3.10.3.1.4 3.10.3.1.1.2
X		Optical plummet	4.5.2.3.8	3.10.3.1.7
X		Level sensitivity	4.5.2.3.1	3.10.3.1.8
X		Tribrach assembly	4.5.2.3.10	3.10.3.1.9
X		Power	4.5.2.6	3.10.3.2
X	X	Accuracy (ELT)	4.5.2.3.2	3.15.2.1
X	X	Measurement time/EDM	4.5.2.4	3.15.2.2
X	X	Range measurement (EDM)	4.5.2.4	3.15.3
X		Data recording unit (DRU)	4.5.2.5	3.10.5
	X	Accuracy (EDM)	4.5.2.4	3.15.3.1
X		Survey system software	4.5.2.5.1	3.16

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4.5.2.2.1 Operational test. The operational test shall be conducted previous to and directly after each of the first article tests indicated in table III. The AISI shall be operated and checked as follows to determine its operational capabilities:

Manipulate all moveable parts and check:	For freedom of movement and non-interference with other parts in various operating positions throughout their maximum ranges.
Fit or attach all accessories to the ELT and check:	For functionality as specified.
Check the levels:	For free movement to determine if the bubble remains within the graduations and if the bubble ends of the coincidence level remain visible.
Rotate the AISI about its horizontal and vertical axes, and check the illumination system:	For uniform and steady operation.
Check parts, components, accessories:	For damage.

Evidence that the AISI components, or accessories will not function as specified, presence of broken, cracked, or shifted parts on the ELT or accessories, or opened joints, loose hardware, or splintered, cracked, or swollen wood on the cases shall constitute failure of this test.

4.5.2.2.2 Pre and post test performance. AISI shall be tested for operational capability (see 4.5.2.2.1) prior to and after each mechanical and environmental test specified in table III. The baseline used for this inspection may be any of the baselines mentioned herein or may be a reference baseline established at the test site. In addition to the baseline, one off-baseline station must be established (minimum angle 45 degrees, minimum distance from AISI 30 meters). The test shall consist of aligning the AISI over the zero monument, taking a distance measurement to another monument, turning the angle to the offset station, and taking an angle and distance measurement and noting all distances, angles and the time to measure each. Accuracy and corresponding time measurements not conforming to 3.15.2.1 and 3.15.2.2 shall constitute failure of the test.

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4.5.2.3 AISI ELT.

4.5.2.3.1 Level sensitivity. The sensitivity of each level vial shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. Evidence that the circular level of the tribrach does not have a sensitivity of 10 minutes, ±3 minutes, shall constitute failure of this test.

4.5.2.3.1.1 Reticle line weight. The reticle line weights shall be determined by using a measuring microscope or by using any other method that will yield comparable values. Evidence of the reticle line weights differing from 2.5 to 4.0 microns shall constitute failure of this test.

4.5.2.3.2 ELT subassembly system tests. Tests shall be reported on a test report form similar to that shown in figure 2 for recording applicable measurements and time. A duplicate of the complete test report form shall be placed within the transport carry case of the instrument to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed as part of the test report form:

This test is based on a series of observations on two collimators approximately 36 degrees apart and on a plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested. This test measures the following parameters of the ELT:

a. Ability of the instrument to repeat measurements.
b. Average circle error.
c. Collimation error.
d. Measurement time.

Two sets of measurements will be made taking into account the special requirements of 3.15.2.2. Each set will consist of 20 angles (40 measurements) by using all sections of the horizontal circle as follows: With the telescope in the direct position, the instrument is pointed on the left collimator with a reading of approximately 0 degrees and 36 minutes. The horizontal angle is read and the seconds are recorded on line 1, column I. The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column I. The telescope is then reversed and pointed on the left collimator, the angle read, and the seconds recorded on line 1, column II (this will be 180 degrees rather than 0 degrees as with the telescope direct). The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column II. The mean of the two readings (columns I and II) is entered in column III. Repeat this technique by changing the position of the horizontal circle to read between 36 and 72 degrees. Advance the circle in this fashion until all parts of the circle are covered during the test.

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The second set of measurements shall be made identically and shall be recorded in columns IV and V. The means of these readings shall be in column VI. Taking the difference between the two mean readings from column III, angle 1 shall be determined and shall be written in column VII. Likewise the difference between two mean readings from column VI shall be made and angle 2 shall be determined and written in column VIII. The difference of these two angles represents the pointing and reading error of the test ($\triangle$). These shall be written in column IX. Column X is $\triangle^2$. From column X, $M_2^2 = \sum \triangle^2/n$ shall be calculated, where n represents the total number of readings. In the case of 36 degrees n = 40, therefore $\sum \triangle^2/40 \leq 4$ sec^2. Numbers for M_2^2 larger than 4 sec^2 indicates excessive pointing and reading error. The test setup should be checked and, after correction, two new complete sets of readings shall be made. Excessive values of M_2^2 the second time indicates that the instrument is not capable of accurately repeating readings and shall constitute failure of this test.

If M_2^2 is smaller than 4 sec^2, proceed with calculations as follows:

Means of the two angles are determined from columns VII and VIII and entered in column XI, mean. These are summed and mean is against determined. Residuals are found and entered in column XII (V). Column XII shall be scanned. Maximum plus and minus deviations shall be determined and underlined. Any value in excess of 2.0 sec shall constitute failure of this test. If plus or minus maximum value is not excessive, compute mean (V). Also compute V^2 and enter in column XIII. Zero is entered opposite the first residual from column XII and the algebraic sum of the residuals is entered in column XIV. These values are added algebraically and a means is determined. This mean is subtracted algebraically from the values of column XIV to give column XV (t). The summation of the column should equal zero. Column XV represents the systematic error of the instrument. Column XVI is t^2. Column XVII is the difference between column I and column II ($D_1 - R_1$). The sum of this column divided by twice the amount of readings equals the collimation error. In case of 36 degrees collimators $C = D_1 - R_1/40 \leq 12$ sec. Any value exceeding 12 sec constitutes failure of this test.

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The circle graduation error t is computed as follows:

$$M_1^2 = \frac{\sum V^2}{n \leq 1} = \frac{\sum V^2}{9}$$

$$M_2^2 = \frac{\sum \Delta^2}{n_R} = \frac{\sum \Delta^2}{40}$$

$$2t^2 = M_1^2 - M_2^2$$

if $M_1^2 - M_2^2 \geq 0$

then $t = \sqrt{\frac{M_1^2 - M_2^2}{2}} \leq 2 \text{ sec}$

Exceeding this value constitutes failure of this test.

if $M_1^2 - M_2^2 < 0$

then compute (t) $= \frac{\sum (t)}{n \leq} = \frac{\sum (t)^2}{10}$

$$(t)\ m = \sqrt{\frac{\sum (t)}{10}}$$

Note: (t) m is not the true circle graduation error; it is the average systematic error that includes the circle graduation error.

Since $M_2^2 \leq 4 \text{ sec}^2$ was earlier established we know that the $t \leq 2$ sec. This is only true if $M_1^2 - M_2^2 < 0$. The number of angles is designated by n. and n_R is the number of readings.

A copy of the completed test data sheets or computer printout of same shall be inclosed with each instrument. Absence of data sheets shall constitute failure of this test.

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4.5.2.3.3 Collimation.

4.5.2.3.3.1 Focus-collimation error. Focus-collimation error shall be determined by observing direct and reversed observations on targets at 4.6 meters and infinity (100 meters) and determined as follows:

a. With the telescope in the direct position, sight on the target in the collimator and read the horizontal angle.
b. Transit the telescope to the reverse position and sight on the target in the collimator and read the horizontal angle. Add 180 degrees to this reading.
c. Subtract the smaller reading from the larger reading, and then divide by 2 and this equals the collimation error.
d. The difference between the error at 4.6 meters and the error at infinity is the focus-collimation error.

Evidence of a focus-collimation error of more than 10 seconds shall constitute failure of this test.

4.5.2.3.4 Telescope.

4.5.2.3.4.1 Effective aperture. A scale graduated in millimeters shall be placed in contact with the front of the objective lens. The effective aperture is the diametrical distance of the lens. An aperture of less than 40 mm shall constitute failure of this test.

4.5.2.3.4.2 Field of view. The field of view shall be measured by pointing the telescope on a collimator or on a target not less than 100 meters from the ELT. The target shall be brought to the intersection of the edge of the field and the horizontal crossline, and the horizontal circle shall be read. The target shall then be brought to the intersection of the horizontal crossline and the opposite edge of the field, and the circle shall again be read. The difference in circle readings is the field of view. The field of view in the vertical plane shall be measured as specified herein, except that the target shall be brought to the intersections of the vertical crossline and the edges of the field and the readings shall be taken from the vertical circle. Evidence of either field of view measuring less than 1 degree 20 minutes for type I instruments or 24 mils for type II instruments shall constitute failure of this test.

4.5.2.3.4.3 Minimum focus. The telescope shall be focused on a collimator, the reticle of which has been set to simulate 2.2 m, or the telescope shall be set at minimum focus and a graduated scale shall be moved along the line of sight until the markings are most sharply defined. The distance from this point of the scale measured along the optical axis to the vertical axis of the instrument is the minimum focus. Evidence of the minimum focus being more than 2.2 m shall constitute failure of this test.

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4.5.2.3.4.4 Resolution. Observations for resolution shall be made within the stadia lines, employing charts contained in NBS Special Publication 374. Resolution of pattern 56 of the high contrast chart, with the chart placed 23.5 meters from the telescope, indicates a resolution of 4 seconds, or better, of arc. Inability of the telescope to resolve the pattern shall constitute failure of this test.

4.5.2.3.4.5 Magnification. The magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light. Without change of focus, a transparent graduated scale shall be placed in front of the objective lens and as near thereto as is practicable. A measuring microscope shall be focused on the image of the scale. The magnification of the telescope shall then be determined in diameters as the ratio of a given length of the scale to the length of its image in the exit pupil. When an autocollimation eyepiece is furnished, the test shall be performed in the same way. Any other method yielding values accurate within 2 percent may be used in lieu of the specified method. Evidence of a magnification for the telescope differing from 28 diameters, +4 or -1 diameter, or of the autocollimation eyepiece having a magnification less than 18 diameters shall constitute failure of this test.

4.5.2.3.5 Reticle lines.

4.5.2.3.5.1 Double vertical reticle lines. With the ELT sighted on a collimator, the interval between the double vertical lines shall be measured by means of the horizontal circle. Readings differing from an apparent angle of 40 seconds, +5 seconds or -10 seconds, shall constitute failure of this test.

4.5.2.3.5.2 Single vertical reticle line. With the ELT sighted on a collimator, the interval between the inner edge of a double vertical reticle line and the adjacent edge of the single vertical reticle line shall be measured. The interval between the inner edge of the other double vertical reticle line and the adjacent edge of the single vertical line shall be measured in the same manner. These angles should be equal. An error of more than ± 0.5 second shall constitute failure of this test.

4.5.2.3.5.3 Stadia lines. With the ELT sighted on a collimator, the length of the horizontal and vertical stadia lines shall be measured by means of the horizontal and vertical circles. Any measurement less than 5 minutes or more than 7 minutes of arc shall constitute failure of this test.

4.5.2.3.6 Vertical circle indexing and vertical collimation.

4.5.2.3.6.1 Altitude level. The collimation error of the vertical circle shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, bringing the altitude level to center, and reading the vertical angle. The operation is repeated in the reverse position. The sum of the readings should equal 360 degrees. Any difference includes twice the vertical collimator error. A net error of more than 12 seconds shall constitute failure of this test.

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4.5.2.3.6.2 Automatic. The collimation error of the vertical circle with a self-indexing device shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, and reading the vertical angle. The operation is repeated in the reverse position. The sum of the readings should equal 360 degrees. Any difference includes twice the vertical collimation error. A net error of more than 12 seconds, or failure of the device to damp within 5 seconds of time shall constitute failure of this test.

4.5.2.3.7 Height of standards. A target placed 30 degrees above the horizon of the instrument and a millimeter scale placed at the same angle below the horizon shall be used to test the standards. The telescope shall be pointed to the target, then tilted downward onto the scale and the reading shall be recorded. The telescope shall be transited and another pointing made. Three sets of direct and reverse readings shall be obtained. The angle formed by the target as apex and the two points where the scale has been read is equal to double the angle of tilt of the trunnion axis with respect to the perpendicular of the vertical axis of the instrument. Evidence of any error more than 12 seconds of arc for the sets of readings after correction for the allowable collimation error shall constitute failure of this test.

4.5.2.3.8 Optical plummet.

4.5.2.3.8.1 Focusing. Failure of the optical system of the plummet to produce a clear image throughout the operating range from tripod height to not less than 33.6 meters shall constitute failure of this test.

4.5.2.3.8.2 Integral with alidade. The optical plummet shall be tested for alignment with the vertical axis by viewing a target and rotating the alidade 360 degrees. Evidence that the line of sight of the optical plummet is not concentric with the vertical axis of the AISI on all points of the circle shall constitute failure of this test.

4.5.2.3.8.3 On tribrach. The AISI shall be centered over a ground marker by the plumb bob. The plumb bob shall be removed and the centering checked by means of the optical plummet. This test shall be repeated with the optical plummet over each side of the tripod head. Evidence that the line of sight is not concentric with the vertical axis shall constitute failure of this test.

4.5.2.3.9 Stadia distance. With the AISI leveled on a stand or tripod, a graduated rod set at a known distance over 30.5 meters from the instrument shall be sighted on. The stadia interval shall be read by means of the stadia hairs, first with the rod in the horizontal position and then in the vertical position. This interval shall be multiplied by 100. Evidence of the stadia distance being in error more than ±0.3 percent of the measured distance shall constitute failure of this test.

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4.5.2.3.9.1 Manual control. When operating on manual control, push

4.5.2.3.10 Tribrach.

4.5.2.3.10.1 Base plate nut. The base plate nut shall be tested for conformance to FED-STD-H28 with certified thread plug gages. Evidence of any base plate nut not mating with the gage shall constitute failure of this test.

4.5.2.3.10.2 Levelment. The ELT shall be mounted on a tripod or on a trivet which will be tilted 6 degrees from the horizontal. The instrument shall be leveled by means of the tribrach foot screws. Any evidence that the instrument cannot be leveled shall constitute failure of this test.

4.5.2.3.10.3 Secure lighting. The AISI will be tested to insure that all external lighting conforms to 3.10.3.1.5.3.

4.5.2.4 AISI EDM accuracy, range, and time. The tests outlined in table III shall be conducted with the AISI EDM in conformance to the requirements of 3.15.3.1 (measurement accuracy), 3.15.3.2 (range), and 3.15.3.3 (measurement time). It is permissible to reject one reading out of the group of 10 required for a distance measurement. More than one reject in a distance measurement shall be cause for rejecting all 10 readings and initiating a new measurement. Rejection of two consecutive distance measurements shall in any case (either reference line or baseline), constitute failure of the test. No allowance shall be made for plumbing errors in test set-ups. The accept/reject criteria herein have already provided for the maximum allowable error. It is permissible to reduce the number of retroreflecting prisms at distances less than 200 meters in order to eliminate possible offset errors and prevent "saturating" the instrument receiver. The AISI shall be tested for accuracy, range, and time in the following set-ups:

a. A baseline consisting of a master monument and a 10-meter beam, graduated at each meter, respectively, with one of the meter spacings further graduated in 10 equal spacings, shall be used. The graduated beam shall be placed with the zero end at a distance of 30 meters from the master monument, then aligned radially away from the master monument. Environmental conditions for this test shall be stable. AISI shall be positioned over the master monument at a height coincident ($\pm$0.10 meter) with the retrodirective prism, mounted on the beam. Distance measurements shall then be made at 0 through 10 meters in 1-meter increments, then 0 through 1 meter, or where graduated, in 0.10-meter increments and corresponding time noted. Accuracy and corresponding time measurements not conforming to 3.15.3.1 and 3.15.3.2 shall constitute failure of the test.

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b. A baseline consisting of a master monument and eight monument stations set on line 100 to 150 meters apart, at the same elevation, ±1.0 meter, with no approximate overall length of 900 meters ±50 meters shall be used. Horizontal baseline distances to first order accuracy shall be used. A distance measurement shall then be made to each station and corresponding time noted. A distance measurement is defined as the average series of 10 readings interrupted only by repointing at the end of the first five readings. Accuracy and corresponding time measurements not conforming to 3.15.3.1 and 3.15.3.2 shall constitute failure of the test.

c. A baseline consisting of a master monument and remote monument located at 2 and 7 kilometers respectively, +200 meters -0 meters, shall be used. AISI shall be positioned over the master monument, a series of three distance measurements taken, and corresponding times noted. Accuracy and corresponding time measurements not conforming to 3.15.3.1 and 3.15.3.2 shall constitute failure of the test.

4.5.2.5 DRU test procedure. The test of the DRU may be conducted in connection with other testing to take advantage of information gathered as a result of other testing. A prepared (on-board) program (e.g., traverse, station establishment, etc.) will be called up and executed in a normal survey task manner. The DRU will be used to collect the data in accordance with manufacturers instructions and the collected data will remain stored in the DRU until completion of all testing. A user defined program for closed traverse by azimuth will be developed and entered in the DRU. A closed traverse by azimuth will be run using the DRU to collect the data (manufacturer's instructions on program development, entry, and use will be followed). Following the manufacturer's instructions the collected traverse data will be transferred from the DRU to a computer system containing a contractor provided processing program. A printout of the raw and processed data will be made for comparison with the original traverse data. The traverse data stored in the DRU will be recalled and arbitrary editing of some data will be performed following manufacturer's instructions. The edited data will be transferred to the computer system and the transferred raw and processed data printed out for comparison with the original traverse and initial data printouts. Following the processing program instructions and the DRU manufacturer's instructions a stakeout routine will be developed and processed on the computer system. Once the data have been developed, those data will be transferred to the DRU (following manufacturer's instructions). The DRU will then be connected to the instrument and the stakeout program executed sufficiently to prove the capability. During all use of the DRU, note will be made of ease or difficulty of operation, reading and relative human engineering factors. Using arbitrary data the traverse program will be expanded until it is no longer possible to enter additional points. The point count for all entered data will be summed to determine total possible point entries. The DRU will be allowed to remain on (under observation) until the operating power warning is evidenced. A new operating power supply shall be installed and the data accessed to determine that no data loss has occurred. As a final check, all data entered in the DRU will be transferred to the computer

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system and printed out for comparison with original data entered. Failure of the DRU to meet the requirements of 3.10.5 through 3.10.5.4 shall constitute failure of this test.

4.5.2.5.1 AISI software evaluation. Software evaluation will be conducted using contractor supplied hardware and with data that have been contractor generated and Government approved. The evaluation will be conducted in a manner that will demonstrate all aspects of the system software. Complete records of all software evaluation will be maintained. The hardcopy output products of the processing actions will be considered as record of the evaluation procedure. The selected software will be loaded on the selected hardware using the procedures provided by the software manufacturer. Problems or difficulties in correctly loading the software will be recorded. Once the software has been loaded and is functioning correctly data, by type, will be entered for processing. Sufficient data will be entered through the data collector to assure its compatibility with the software. For the topographic (control) application evaluation, the data necessary to perform the functions listed in 3.16 will be input in accordance with criteria contained in this paragraph. Hardcopy output will include raw data, edited data, all files developed, and the processed data to demonstrate compliance with the topographic function requirements listed in 3.16. In the case of construction applications, the data used for topographic evaluation may be used where applicable; however, any additional data necessary to demonstrate conformance with the construction requirements listed in 3.16 will be entered. Hardcopy output will include raw data, edited data, all files developed, and the processed data necessary to demonstrate compliance with the construction survey requirements listed in 3.16.

4.5.2.6 Power.

4.5.2.6.1 Stability. AISI shall be tested for stability by aligning over one of the baselines, then with a fully charged internal battery take readings continuously until the battery is expended or the low voltage cutoff is reached as noted on the battery strength meter. Accuracy measurements not conforming to 3.15.3.1 and 3.15.3.2 shall constitute failure of this test.

4.5.2.6.2 Power consumption. While AISI is being tested in 4.5.2.6.1, power shall be continuously monitored. Power consumption in excess of that specified in 3.10.4.1 and 3.10.3.2 shall constitute failure of this test.

4.5.2.6.3 Battery life.

4.5.2.6.3.1 Main power source. AISI shall be set up on a baseline used in 4.5.2.2.2 and measure the distance using the main power source (NiCad battery), until the power is insufficient for operation of the AISI or the accuracy of the measurements are not in compliance with 3.15.3.1 and 3.15.3.2. The total operating time shall be determined and nonconformance to 3.10.3.2.1 shall constitute failure of this test.

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4.5.2.6.3.2 External power source. AISI shall be set up on a baseline used in 4.5.2.2.2 and measure the distance using the external power source (lead acid battery). Nonconformance to 3.10.3.2.2 and 3.10.4.3 shall constitute failure of this test.

4.5.2.7 Environmental.

4.5.2.7.1 High-temperature and low-temperature operation.

4.5.2.7.1.1 Test chamber. The high- and low-temperature operation tests shall be conducted in a chamber of a size that will house AISI, mounted on its tripod, and accessories and shall permit the inspector to walk around the periphery of the tripod legs in the course of the examination and tests specified herein. Temperature control tolerance for the test chamber shall be ± 3 °C or as noted, for all tests specified herein, and humidity control tolerance shall be ± 5 percent for all tests specified herein. All temperature gradients shall be 14 °C/hr ± 3 °C.

4.5.2.7.1.2 High-temperature operation. AISI, mounted in the operating position (with cases, battery, and all accessories) shall be placed in a temperature-controlled environmental test chamber (see 4.5.2.7.1.1). The temperature in the chamber shall be gradually raised at 14 °C/hr to 50 °C, +3 °C, -0 °C, and maintained at that temperature for not less than 4 hours. Temperature readings shall be continuously maintained. Repeat process for 4 cycles. Evidence of damage (see 6.5) to any AISI hardware, failure of AISI componentry to operate, or nonconformance to 4.5.2.2.2 shall constitute failure of this test.

4.5.2.7.1.3 High-temperature storage. Upon completion of high-temperature operational testing (see 4.5.2.7.1.2), the AISI mounted in the operating position shall be subjected to not less than 71 °C and held at that level for not less than 4 hours. At the end of the 4-hour period, the temperature shall be lowered to ambient and AISI shall be subjected to the performance test specified in 4.5.2.2.2 and all accessories shall be examined for damage. Evidence of damage (see 6.5), inability of any accessory to function properly, or nonconformance to 4.5.2.2.2 shall constitute failure of this test.

4.5.2.7.1.4 Solar radiation. AISI, mounted in the operating position, shall be stabilized in the test chamber (see 4.5.2.7.1.1) at 43 °C, and a source of directed radiant energy simulating solar radiation of not less than 355 Btu per square foot per hour shall be applied to AISI for a period of not less than 4 hours. At the end of this period, AISI shall be subjected to the performance test specified in 4.5.2.2.2. Evidence of damage, inability of any accessory to function, or nonconformance to 4.5.2.2.2 shall constitute failure of this test.

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4.5.2.7.1.5 Low-temperature operation. AISI, mounted in the operating position (with cases, battery, and all accessories) shall be placed in a temperature-controlled environmental test chamber (see 4.5.2.7.1.1). The temperature in the chamber shall be gradually lowered 14 °C/hr to -20 °C, +0 °C, -3 °C, and held at that level for not less than 4 hours. AISI shall then be subjected to the performance test specified in 4.5.2.2.2 and inspected for damage (see 6.5.1). When applicable, the winterization kit shall then be installed on AISI, batteries removed from the test chamber and the chamber temperature lowered to -32 °C, +0 °C, -3 °C, and held for not less than 4 hours. The winterization kit shall be energized. After a 25-minute warming cycle, the AISI battery shall be attached; then AISI shall be inspected and tested as specified in 4.5.2.2.2. The AISI battery shall again be removed and the chamber temperature lowered to -43 °C and held for not less than 4 hours. The winterization kit shall be energized. After a warming period of not more than 45 minutes, the AISI battery shall be attached; then AISI shall be inspected and tested as specified in 4.5.2.2.2. Evidence of damage (see 6.5.1) or nonconformance to 4.5.2.2.2 shall constitute failure of these tests.

4.5.2.7.1.6 Low-temperature storage. Upon completion of low-temperature operational testing, (see 4.5.2.7.1.5) the chamber temperature shall be lowered to -45 °C and held for a period of not less than 6 hours. At the end of the 6-hour storage period, the chamber temperature shall be returned to laboratory conditions, allowed to stabilize and dry out for a period of not less than 24 hours at a temperature not less than 29 °C. AISI shall then be inspected for damage (see 6.5.1) and tested as specified in 4.5.2.2.2. Evidence of damage (see 6.5.1) or nonconformance to 4.5.2.2.2 shall constitute failure of the test.

4.5.2.7.2 Humidity. Place the AISI, with power off, mounted in the operating position, with its battery, open cases and all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2. The chamber temperature shall be raised to 49 ±1 °C, and 95 percent ±2 percent humidity applied. These conditions shall continue for not less than 24 hours at which time the temperature and humidity shall be removed and conditions allowed to stabilize for 4 hours at ambient conditions. AISI shall be inspected and tested as specified in 4.5.2.2.2. Evidence of damage (see 6.5.1) or nonconformance to 4.5.2.2.2 shall constitute failure of this test.

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4.5.2.7.3 Rainfall. AISI, mounted in the operating position, with its battery and with cases closed and containing accessories, shall be subjected to rain in accordance with MIL-STD-810, method 506.2, and the following procedure:

The wind source shall be installed and adjusted to produce a horizontal velocity of not less than 48 km/hr. The test item shall be exposed to simulated wind-driven rainfall at a rate of 7.6 cm ±2.5 cm/hr for not less than 5 minutes. At the end of the 5-minute exposure, install the rain covers, as applicable. The wind shall be maintained at this velocity for not less than 55 minutes. Three of the four sides of the instrument and cases shall be subjected to not less than 15 minutes of wind-driven rain, except the first side, which shall be exposed for not less than 10 minutes. The total test duration, then, is not less than 1 hour. At the end of 1 hour, the rain and wind shall be removed. AISI instrument, battery, and cases shall be wiped dry. The protective cover(s), when applicable, shall then be removed and the test item inspected, with particular attention to evidence of water penetration, such as free water, swelling of material or other deterioration in the instrument of cases, and tested as specified in 4.5.2.2.2. Evidence of damage (see 6.5.1), water penetration or nonconformance to 4.5.2.2.2 shall constitute failure of the test. Water found inside AISI which is deemed to have entered during post rain inspection or during removal of covers shall not be considered.

4.5.2.7.4 Shock.

4.5.2.7.4.1 AISI in transport case. AISI, stowed in transport case(s), shall be shock tested in accordance with MIL-STD-810, method 516.3, procedure II, except that the drop height shall be not less than 91 cm. There shall be one flat drop on each of the six faces, and four corner drops. Corners selected for shock shall be optional; however, two corner drops shall be sustained by the bottom shell of the case and the other two sustained by the top shell of the case. In all drops, the case shall be restrained from the secondary shock by bouncing or tumbling. At the conclusion of this shock test, AISI shall be inspected for damage and tested as specified in 4.5.2.2.2. Evidence of damage (see 6.5.1) to AISI componentry, excluding case(s) or nonconformance to 4.5.2.2.2 shall constitute failure of the test. AISI case(s) shall be examined. Inability of the case(s) to protect AISI componentry shall constitute failure of the test. Damage to case(s) shall be allowed provided that it does not prohibit the case from continuing its mission. Examples of allowable damage are as follows:

a. Superficial hairline cracks.
b. Deformation of sealing edges less than 0.32 cm end-to-end of case which does not inhibit action, leveling action, or sealing.
c. Wrinkling of corners which will rebound.
d. Locking latches distorted but not loose at mountings or allowing case seal to deteriorate and still able to function.
e. Handles abraded but not deformed.
f. Case abrasion but no through-the-wall rupture.

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4.5.2.7.5 Vibration.

4.5.2.7.5.1 AISI in transport case. The vibration test for the instrument and accessories in the transport case(s) shall be performed with the case(s) rigidly mounted to the vibration table. The test shall be conducted in accordance with MIL-STD-810, method 514.3 with the following exceptions:

a. One 5-minute logarithmic cycle for each of the three major axes, from 7 to 200 Hz and return, with a maximum acceleration of 1.5 g, shall be performed.
b. The transport case(s) shall be removed from the vibration table and the test specified in 4.5.2.2.2 shall be performed. Nonconformance to 4.5.2.2.2 or evidence of damage shall constitute failure of this test.

4.5.2.7.6 Magnetic environment. AISI in the following operating position (internal battery and applicable accessories) shall be positioned over a point from which a baseline distance has been established. A magnetic source shall be positioned to produce a magnetic field of not less than 3 gauss (G) at the instrument. Ten distance readings shall be made with the lines of magnetic force directed in each of the following directions: North-south, east-west, and vertical. Readings and corresponding times not conforming to 3.4.8 shall constitute failure of the test. The test will be repeated using an external battery source and applicable accessories. Nonconformance to 3.4.8 shall constitute test failure.

4.5.2.7.7 Electromagnetic interference. The AISI in the following operating position (internal battery and applicable accessories) shall be placed, on a stand 76 cm ±5 cm above the facility floor. No external ground shall be attached. The instrument's external power source with external power cable (battery pack) shall be positioned on the stand beside the instrument during both the radiated and conducted tests. Antenna equipment shall be oriented at 1 meter test distance from the instrument with all emissions being measured from one position. The current probe (detecting device) shall be attached to the input power leads to the instrument to measure the conductive emissions. In the event that the input power leads are separated, the conductive emissions shall be measured on each lead individually. If the leads are bonded together as a single power cable, both leads shall be measured together. Instruments having interconnecting leads other than power source leads shall also be measured for conductive emissions. Nonconformance to 3.9 shall constitute failure of this test. The test will be repeated using an external battery source and applicable accessories.

4.5.2.7.8 Sand and dust. The AISI, not in its transport case(s), without its lens cover, and the closed transport case(s) shall be placed in a sand and dust testing chamber not less than 91 cm high, 91 cm wide, and 91 cm deep. Sand and dust of 140 mesh silica flour shall pass through the chamber at a velocity of 30 to 150 meters per minute. The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature

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shall be maintained at 25 ±3 °C. The relative humidity shall not exceed 30 percent. The AISI shall be subjected to this test for a period of 15 minutes. Nonconformance to 3.4.9 shall constitute failure of this test.

4.5.2.8 Human factors engineering. AISI shall be tested for human engineering factors as required in 3.7. During the course of all testing herein, test directors shall scrutinize AISI to determine hardware compatibility to required operator functions. The following functions shall be noted specifically:

a. Suitability for backpack.
b. Operation using arctic mittens.
c. Ease of sighting, tangent running, and clamping.
d. Adaptability to night operation.
e. Location and configuration of all controls.
f. Compatibility of operation with other surveying hardware.
g. Adaptability to charging battery.
h. Interchangeability of AISI componentry with other hardware during operation.
i. Adaptability to easy "setup" in field operation.
j. Hardware presents no safety hazard from electronics, optics, or mechanical hazard to operator, maintenance personnel, and others.
k. Operation using chemical/biological protective clothing.
l. Easily and quickly aligned (bore-sighted) in the field.

4.5.2.9 Safety. The AISI contractor shall provide the Government contracting officer certified test laboratory data (see 6.6) substantiating the AISI meets the criteria cited in 3.6. Concurrently, all AISI radiometric data shall be sent to the Army Environmental Hygiene Agency, Aberdeen Proving Ground, Maryland.

4.5.2.10 Reliability. The AISI shall be tested for reliability. The test values shall be as follows:

Upper θ_0 = 240 Missions
Lower θ_1 = 120 Missions

The reliability test shall consist of the following operational profile repeated 450 times for a total of 450 missions. (A mission is defined as the performance of steps a. through d. indicated below.)

Experiencing three or more failures shall be cause for rejection of the unit. Equipment at the start of this test shall be packed in transport cases.

a. Unpack and set-up equipment for operation over certified baseline with targets of known angular value observable.

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b. Perform ten distance and six horizontal angle measurements to selected targets within the maximum range limits of type AISI being tested. Angular measurements shall consist of six pointings in the direct telescope position and six in the reverse telescope position. Measurement accuracies shall conform to requirements in 3.15.2 and 3.15.3. All mission data shall be transferred and stored in the DRU. Upon completion of the test, all data stored in the DRU shall be retrieved and verified for accuracy.
c. Dismantle and repack equipment for transport.
d. Perform manual transport of not less than 2 meters and return to original position.

Note. The use of a pedestal or detachment of instrument tribrach at the occupied station is allowable to minimize plumbing errors. Power for the reliability test shall be supplied by either appropriate internal or external sources.

4.5.2.10.1 Failure definition. A failure shall be defined as any malfunction, degradation of performance, or existence of personnel or equipment safety hazard.

4.5.2.10.1.1 Relevant failures. All failures shall be classified as relevant except as specified in a. through g.:

a. Installation damage.
b. Accident or mishandling.
c. Failure of the test facility or test-peculiar instrumentation.
d. Equipment failures caused by an externally applied overstress condition, in excess of the approved test requirements.
e. Normal operating adjustments (nonfailure) specified in the approved equipment operating instruction.
f. Secondary failures within the equipment, which are directly caused by nonrelevant or relevant primary failures. The secondary failures must be proved to be dependent on the primary failure.
g. Failures caused by human errors.

4.5.2.10.1.2 Chargeable failures. All relevant failures requiring more than 30 minutes total operator corrective time to continue with the mission shall be charged as a mission reliability failure.

4.5.2.11 Maintainability.

4.5.2.11.1 Maintenance support evaluation. Maintenance operation required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support equipment to maintain the AISI shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance support.

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4.5.2.11.2 Maintenance ratio. The maintenance ratio shall be computed during testing specified in 4.5.2.10. Number of missions and the active maintenance man-hours required for both scheduled and unscheduled maintenance shall be recorded. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product consisting of the whole AISI equipment system.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

Major

130. Desiccant not placed in the case as specified for level A, B, or C.
131. Quantity of desiccant not in accordance with the referenced table for level A and B (see 5.1.1).
132. Wood block not secured in end of box containing the tripod as specified for level A or B.
133. Fiberboard boxes not closed and secured in accordance with the referenced methods as specified for level A or B.
134. Shipping containers not as specified.
135. Shipping containers not lined with barrier material as specified for levels A, B, or C.
136. Barrier material not as specified for levels A, B, or C.
137. Cushioning material not as specified for levels A, B, or C.
138. Marking missing, illegible, incorrect, or incomplete (see 5.3).

Minor

215. Strapping not zinc coated for level A (see 5.2.1).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. All components for which cases are provided shall be placed in their respective cases and the case lids closed and secured. In addition, the case(s) containing the AISI shall have desiccant placed within the case. The

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desiccant shall conform to MIL-D-3464 and shall be in the quantity required in accordance with the formula specified in MIL-P-116. The components shall be placed in the transport case as provided for and the case lid closed and secured. Components such as tripods, for which cases are not provided, shall be placed in close-fitting boxes conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade W5c. The box containing the tripod shall have a fitted block of wood secured in the end of the box to prevent the tripod legs from puncturing the box. Each box shall be closed and secured in accordance with method V of the appendix to the box specification.

5.1.2 Level B. Preservation shall be as specified for level A, except that boxes for components shall be type CF, class domestic, variety SW, grade 175. Box closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. All components shall be preserved as specified for level B, except the components without cases shall be placed in close-fitting boxes that will prevent damage to the contents.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The components comprising a complete AISI, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Each box shall be lined with barrier material conforming to MIL-B-81705 to prevent electrostatic damage to the electrical items. Box closure and strapping shall be in accordance with tl appendix to the box specification, except strapping shall be flat and finish B. Cushioning material shall conform to PPP-C-795, class 2 or PPP-C-1842, type III, style A.

5.2.2 Level B. The components comprising a complete AISI, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type and strapping may be finish A.

5.2.3 Level C. Each complete AISI, preserved as specified in 5.1, shall be packed together in a close-fitting wooden box that will prevent damage to the contents during shipment. Each box shall be lined with barrier material conforming to MIL-B-81705 to prevent electrostatic damage. Cushioning material shall conform to PPP-C-795, class 2 or PPP-C-1842, type III, style A.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The AISI shall be used to determine distances and directions for; field artillery weapons systems, establishing geodetic survey control and construction surveys.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type AISI required (see 1.2).
c. Part number (see 1.2.1).
d. Date of issue of DODISS applicable and exceptions thereto (see 2.1.1).
e. When a first article is not required for inspection and approval (see 3.2).
f. When a prism adapter is required (see 3.10.6).
g. When a winterization kit, with thermometer conforming to GG-T-355, altimeter/barometer conforming to A-A-52005, and meteorological correction charts (see 3.10.9), is required.
h. When a backpack is required (see 3.10.9.1).
i. When tripods are required (see 3.10.9.3).
j. Color when other than as specified (see 3.13).
k. Only one survey system software package per every three AISIs (see 3.16).
l. Level of preservation and packing required (see 5.1 and 5.2).
m. Any special marking (see 5.3).

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, including installation procedures, instructional materials, illustrated parts lists, early delivery of AISI systems for training development, and contractor's maintenance and operational manual to be furnished with each instrument.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of instruments.

6.5 Definition.

6.5.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the AISI with provisions of this specification.

6.6 Safety. The contracting officer should arrange to furnish to the contractor a copy of TB MED 524 and MIL-STD-454 and shall require the contractor to furnish certified radiometric data of the infrared radiation output of a representative surveying instrument of his own manufacture that complies with this specification to the contracting officer or his technical representative in accordance with AR 40-5 for the purpose of determining that all applicable safety requirements in 3.6 have been complied with as specified in 4.5.2.9. The contracting officer shall provide the contractor with a list of certified test laboratories authorized to perform required safety testing and measurement.

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6.7 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, tests and approval of the first article test results and disposition of the first article.

6.8 Corrosion-resisting materials. Corrosion-resisting steel, which contains a minimum of 12 percent chromium, copper, brass, aluminum, and aluminum alloys which contain not more than 0.40 percent copper, is considered a corrosion-resisting material.

6.9 Subject-term (key word) listing.

a. Total station system.
b. Surveying system.
c. Electronic distance meter.
d. Electronic theodolite.
e. Data recording unit.
f. Automated integrated survey instrument.

Custodians:
Army - ME

User activity:
Navy - YD

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-0845

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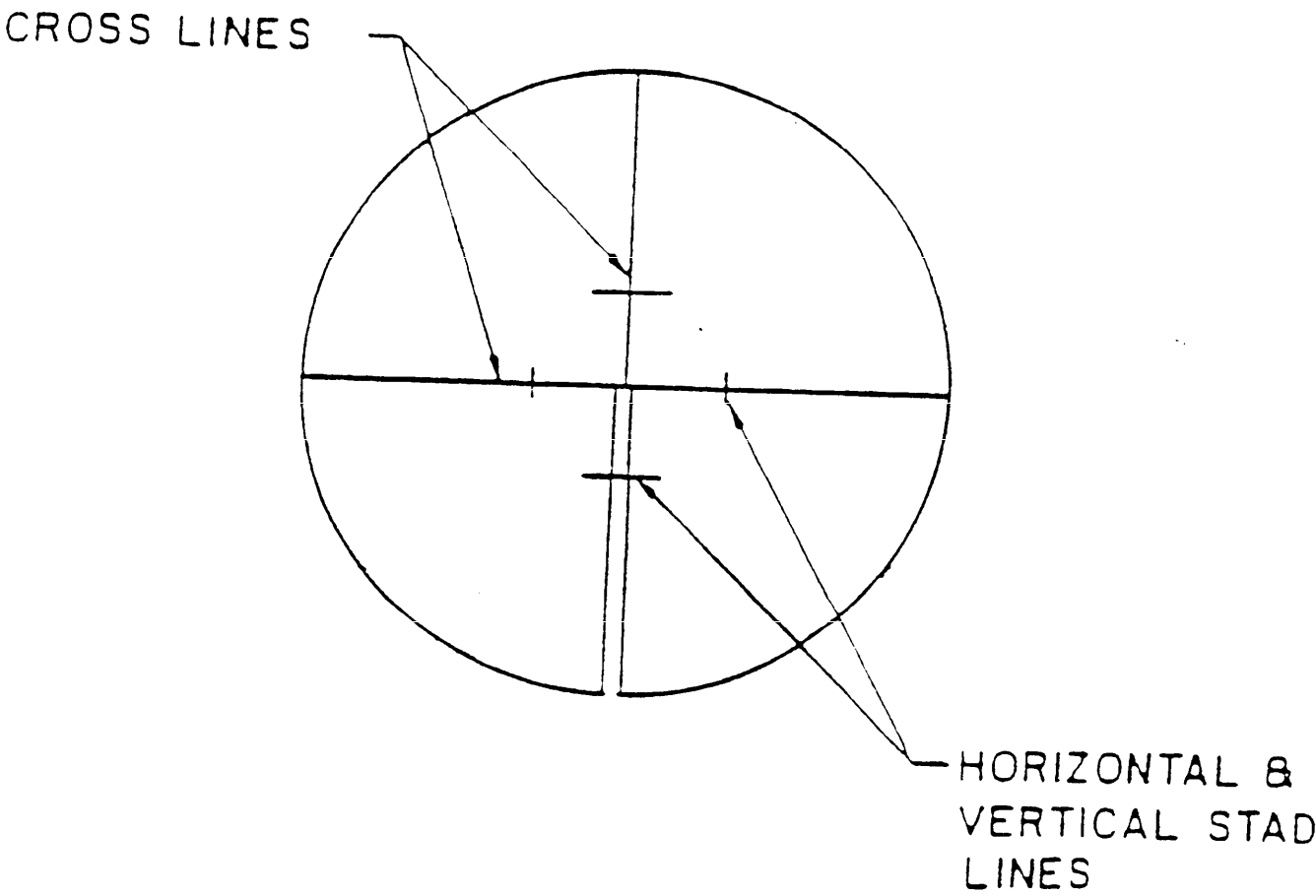


Figure 1. Reticle pattern for theodolite.
(illustrative example only)

X-4785

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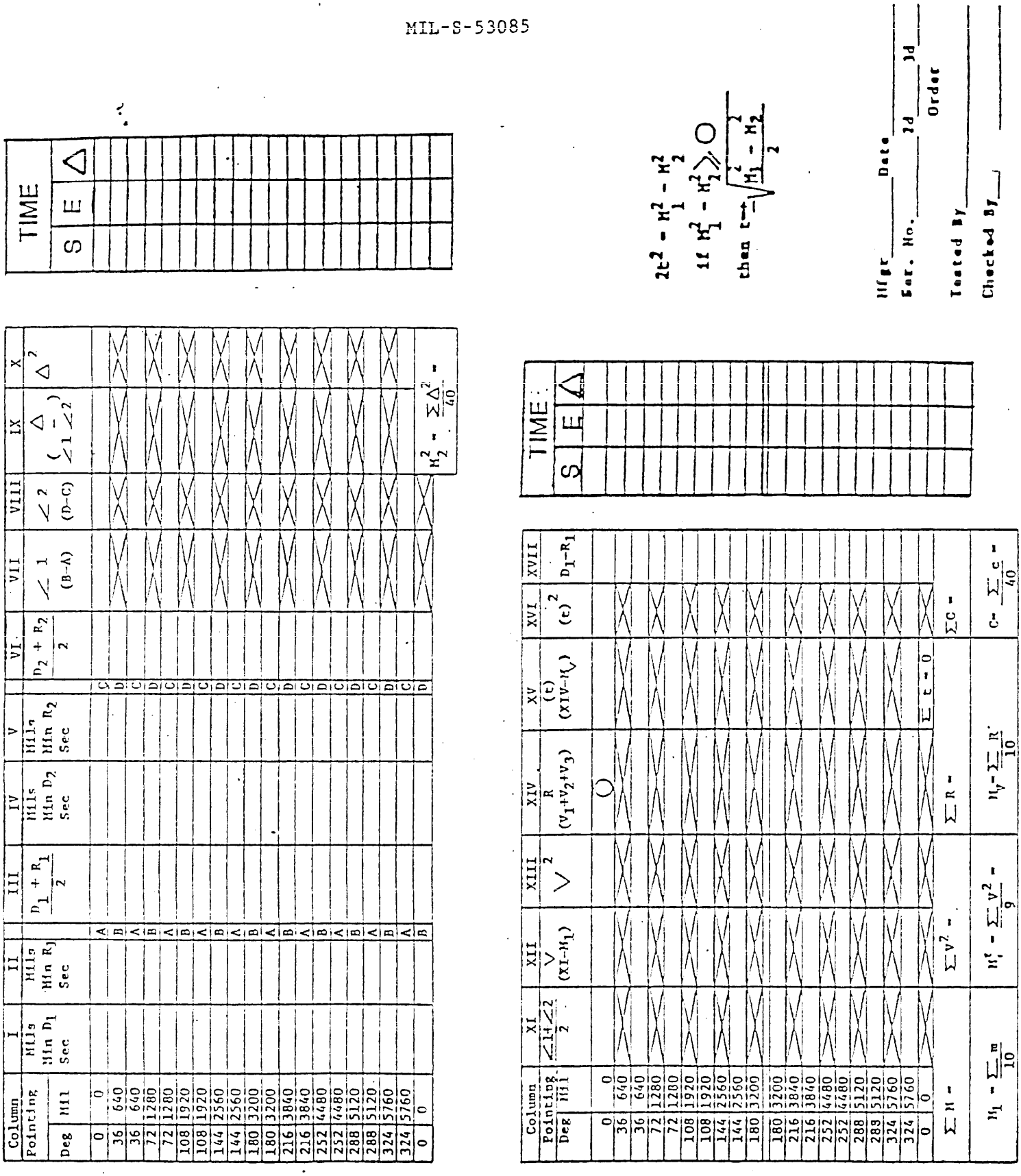


Figure 2. Data sheet for theodolite system test.
(horizontal)

X-4786

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-53085	Survey Instrument Automated, Integrated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREA

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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